

Employee Management System

Project Overview

The Employee Management System is a console-based Java application developed to manage employee records. The application allows users to perform CRUD (Create, Read, Update, Delete) operations on employee data through a menu-driven interface.

Technologies Used

- Java
- Object-Oriented Programming (OOP)
- Collections Framework (ArrayList)
- Exception Handling
- Scanner Class

Project Structure

Employee.java

Contains employee details:

- Employee ID
- Employee Name
- Salary
- Department

Implemented:

- Private fields
- Constructors
- Getters and Setters
- toString() method

EmployeeService.java

Contains business logic for employee operations:

- Add Employee
- View Employees
- Search Employee
- Update Employee
- Delete Employee

Used ArrayList to store employee records.

EmployeeManagementSystem.java

Main class containing:

- Menu-driven program
- User interaction
- Employee service method calls
- Exception handling

Features Implemented

1. Add Employee

- Accepts employee details from the user.
- Stores employee information in ArrayList.
- Prevents duplicate employee IDs.

2. View Employees

- Displays all employee records.
- Shows a message when no employees are available.

3. Search Employee

- Searches employee by ID.
- Displays employee details if found.
- Displays "Employee Not Found" if not found.

4. Update Employee

- Updates employee name, salary, and department using employee ID.

5. Delete Employee

- Deletes employee record using employee ID.

6. Exception Handling

Implemented InputMismatchException handling to prevent application crashes when invalid data types are entered.

Example:

- Entering a name instead of salary.
- Entering characters instead of numeric menu choices.

The application displays:

"Invalid Input! Please enter the correct data type."

and continues execution without terminating.

OOP Concepts Used

- Class and Object
- Encapsulation
- Constructors
- Method Invocation
- Data Abstraction through Service Layer

Learning Outcomes

Through this project, the following concepts were practiced:

- Core Java Programming
- Object-Oriented Programming
- Collections Framework
- Exception Handling
- Menu-Driven Application Development
- CRUD Operations

Future Enhancements

- MySQL Database Integration
- JDBC Connectivity
- Spring Boot Integration
- REST API Development
- Spring Data JPA
- Hibernate
- Web-Based User Interface

Conclusion

This project successfully demonstrates the implementation of an Employee Management System using Core Java. It provides CRUD functionality, exception handling, and proper object-oriented design principles in a console-based application.